STUDENT PRINT PACKET

EBIKE101 Day 3

Motors, Controllers, Throttle/PAS, and Signal Flow

Trace the system before you test the system.

Name: _____ Team: _____

Today's question

How does rider input become controlled wheel motion, and where can that chain fail?

Core rule

Power off first. Trace before testing. Control motion before adding power.

Morning Quiz

Name: _____ Team: _____

Choose the best answer. If unsure, choose the safest professional action.

1. What is the first action before tracing an unknown e-bike circuit?

A. Add throttle
B. Make the system safe and power it off
C. Disconnect random plugs
D. Replace the motor

2. What is the controller's main job?

A. Inflate the tires
B. Store all battery energy
C. Read inputs and manage power sent to the motor
D. Hold the rear wheel

3. What does the motor do?

A. Converts electrical energy into mechanical rotation
B. Measures tire pressure
C. Charges the battery
D. Creates a brake signal

4. What does a throttle normally provide?

A. A direct request for motor assist
B. Mechanical braking
C. Battery cooling
D. Wheel alignment

5. What does PAS detect?

A. Tire pressure
B. Brake pad thickness
C. Pedaling input
D. Battery swelling

6. What should a brake cutoff do when the brake is applied?

A. Increase assist
B. Tell the controller to stop motor drive
C. Charge faster
D. Ignore the rider

7. Which sequence is the power path?

A. Battery -> protection -> controller -> motor
B. Throttle -> display -> helmet -> tire
C. Brake -> chain -> charger -> motor
D. PAS -> seat -> controller -> wheel

8. Which parts mainly send signals to the controller?

A. Frame, fork, and tires
B. Chain, pedals, and spokes
C. Throttle, PAS, brakes, and display
D. Saddle, grips, and fenders

9. What is the difference between signal and power?

A. They are always the same wire

B. Signal carries a command; power carries energy
C. Power carries only data
D. Signal turns the wheel by itself

10. A bike has no assist. What comes before replacing parts?

A. Order a new motor
B. Bypass every cutoff
C. Make safe and confirm the complaint
D. Force each connector

11. What is the safe response to an unknown connector?

A. Force it until it seats
B. Identify and verify it before connection
C. Cut off the keying tab
D. Try every matching color

12. Where should wiring be routed?

A. Clear of moving parts, sharp edges, heat, and pinch points
B. Against the tire
C. Around the crank
D. Across the brake rotor

13. Before an instructor-controlled powered wheel test, the bike should be:

A. Held by one student
B. Pointed at a doorway
C. Stable with the driven wheel and area controlled
D. In the highest assist mode

14. Where does a hub motor apply torque?

A. At the handlebar
B. At the wheel hub
C. Through the seatpost
D. Inside the brake lever

15. Where does a mid-drive apply torque?

A. Through the bicycle drivetrain
B. Only through the front brake
C. Through tire pressure
D. Directly through the saddle

16. What makes a diagnostic result useful to the next person?

A. A guess
B. A replaced part with no test
C. A recorded checkpoint and result
D. A faster throttle test

17. Unexpected motor motion means:

A. Hold the wheel harder
B. Stop-work and make the system safe
C. Increase assist
D. Continue if it is brief

18. What should be checked when identifying a connector?

A. Shape, keying, pin count, labels, and routing
B. Only wire color

C. Only connector size
D. Only whether it fits with force

19. Which conditions come before a powered test?

A. Full speed and a camera
B. Only a charged battery
C. Instructor approval, stable bike, clear area, and controlled motion
D. Removing the brakes

20. What is today's core diagnostic lesson?

A. The motor is always the problem
B. Trace the system instead of guessing at parts
C. Every connector should be unplugged
D. Powered tests come first

Part 1: Component map

Find each part on the bike or component table. Describe its job and mark whether it belongs mainly to the signal path, power path, or both.

Component	Job	Signal / Power / Both
Battery		
Protection/fuse		
Controller		
Motor		
Display		
Throttle		
PAS		
Brake cutoff		
Main connectors		

Part 2: Draw both paths

Power path: Battery -> _____ -> _____ -> Motor -> _____

Signal path: Throttle / PAS / Brakes / Display -> _____ -> controlled output

Where do the paths meet? _____

Why must a brake signal override a throttle or PAS request? _____
_____

Afternoon Lab: Control-path tracing

Safety boundary

- Battery remains off or disconnected during component tracing.
- Do not force, guess, back-probe, or disconnect an energized connector.
- No powered wheel test without instructor approval, a stable bike, and a clear area.
- Unexpected motion, damaged wiring, exposed conductor, heat, swelling, odor, or unknown polarity means stop-work.

Team roles

Role	Job
Tracer	Follows the physical path
Connector inspector	Checks shape, keying, pins, labels, and routing
Diagrammer	Draws signal and power paths
Reporter	Explains evidence and the first safe test

Lab A: Identify

- Stabilize the bike and confirm power is off.
- Tag the battery, controller, motor, display, throttle/PAS, brake cutoff, and major connectors.
- Mark each part S for signal, P for power, or B for both.
- Record any unknown part instead of guessing.

Lab B: Trace

- Trace the power path from battery to motor.
- Trace the signal path from rider and safety inputs to the controller.
- Mark the point where the two paths meet.
- Circle every connector or checkpoint that could cause a no-assist complaint.

Lab C: No-assist diagnostic plan

Instructor-selected complaint or fault card: _____

Step	Checkpoint	Expected evidence	Actual result
1	Make safe	Bike stable, power state known	
2	Confirm complaint	Exact condition stated	
3	Power source	Battery and protection status verified	
4	Display / wake-up	System command state known	
5	Safety inputs	Brake cutoffs not blocking assist	
6	Rider input	Throttle or PAS request identified	
7	Controller path	Inputs and outputs traced	
8	Motor path	Cable and connection inspected	

First safe test

What would you test first, and why?

_____

_____

Exit ticket

- Explain the difference between a signal path and a power path.
- Name two safety inputs that can stop or limit assist.
- What evidence would stop you from performing a powered test?
- What is the first safe checkpoint in your no-assist plan?